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infDB: An Open-Source Data Ecosystem for Urban Energy Infrastructure Modelling and Planning

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Summary

`infDB` (Infrastructure and Energy Database) is a modular, open-source data infrastructure designed to streamline energy modeling and municipal heat planning. It reduces the overhead of data collection and pre-processing by offering a containerized platform for standardized data ingestion, storage, and retrieval. By providing a unified interface for heterogeneous data sources, `infDB` allows researchers and planners to focus on generating insights and modeling energy systems rather than managing data logistics. The platform fosters a collaborative ecosystem by ensuring data is FAIR (Findable, Accessible, Interoperable, and Reusable).

Statement of need

The transition to climate-neutral heating is a central pillar of energy policy, exemplified by Germany's aim for climate neutrality by 2045. New legislative frameworks, such as the requirement for Municipal Heat Planning (KWP) and the transparency obligations of the German Energy Industry Act (EnWG §14d), demand that municipalities and Distribution System Operators (DSOs) process vast amounts of infrastructure data (*BMWSB Kommunale wärmeplanung - BMWSB*, 1/19/2026; *EnWG §14d*, 1/19/2026).

However, the current landscape of energy data is fragmented. While the Open Data Strategy of the Federal German Government (*Data Strategy of the Federal German Government*, 1/27/2021) has increased data availability, this data is published by disparate authorities in varying formats, spatial resolutions, and licensing structures (e.g., CC-BY 2.0). Consequently, energy modeling workflows often suffer from:

1. **Lack of Reproducibility:** Heat plans are often one-off studies that are difficult to update or audit.
2. **High Pre-processing Effort:** Researchers spend disproportionate time standardizing raw data before analysis.
3. **Siloed Infrastructure:** DSOs and municipalities lack standardized tools to manage and share distribution network data efficiently.

`infDB` addresses these challenges by providing a reproducible, version-controlled, and automated ETL (Extract, Transform, Load) pipeline. It acts as a middleware between raw public data and high-level energy modeling tools, ensuring that planning data is transparent, traceable, and easily updatable.

State of the field

Energy and infrastructure data management is an active field with several existing solutions:

- **Commercial Platforms:** Tools like **nPro**, **Solarea**, and **flexRM** offer robust analytics and user interfaces but are proprietary, limiting transparency and community extension.
- **Open Source Modeling Frameworks:** Tools like **City Energy Analyst (CEA)** and **OpenPlan** excel at simulation and optimization but often assume the existence of cleaned, structured input data.
- **Data Initiatives:** The **NEED project** (*NEED - Neue Daten Für Die Energiewende*, 9/01/2023) provides a decentralized data hub for synthetic energy data, and **DB4KWP** (*Vom Plan Zur Wende*, 1/19/2026) focuses on ontologies (OEO/OEKG) and naming conventions.

`infDB` addresses a critical gap in the energy data ecosystem. While simulation tools like the City Energy Analyst excel at modeling, `infDB` provides the foundational data infrastructure that these tools require. Unlike static data repositories, `infDB` offers an active, containerized platform that enables users to deploy local instances, continuously integrate fresh datasets, and seamlessly connect with both commercial and open-source downstream tools. By providing a technical implementation layer, `infDB` can complement ontology initiatives like DB4KWP, transforming conceptual data standards into practical, operational systems.

Software Design

`infDB` follows a modular microservices architecture orchestrated via Docker Compose. The system is conceptually divided into **Services**, which provide the foundational infrastructure, and **Tools**, which consume and process the data. This separation allows for high portability and enables users to activate only the components required for their specific use case.

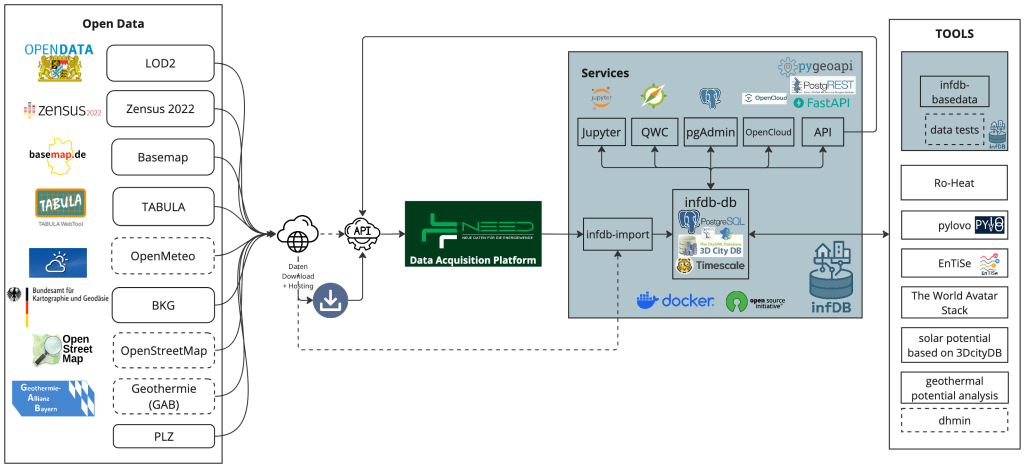

Figure 1: infDB - Services and Tools

The *Services* layer (depicted in the grey box in the architecture diagram) handles database operations, administration, data ingestion, and connectivity. These containerized services include:

- **infdb-importer:** This service automates the ingestion of diverse open data sources. It transforms raw external formats into structured schemas within the database. Users control this process via a simple YAML configuration file (`config-infdb-loader.yml`), eliminating the need for custom ETL scripting.

- **infdb-db:** The central storage engine hosting a PostgreSQL database. It is pre-configured with essential extensions for energy modeling:
  - **PostGIS** for geospatial data.
  - **TimescaleDB** for time-series data.
  - **3D City DB** for city model data.
  - **pgRouting** for graph-based network analysis.
- **APIs:** This layer provides standardized interfaces to external applications, ensuring the database remains accessible but secure. It includes **FastAPI** for custom logic, **pygeoAPI** for OGC standards, and **PostgREST** for immediate RESTful access to database tables.
- **pgAdmin:** A web-based GUI for database administration, allowing users to inspect schemas, run queries, and manage data without command-line interaction.
- **JupyterNotebook:** An interactive environment pre-loaded with the `pyinfdb` library, enabling users to prototype data analyses and visualize results directly within the platform.
- **QGIS Webclient:** A web-based GIS client for visualizing geospatial data stored in the database, useful for urban planners and GIS specialists.
- **OpenCloud:** An optional component for integrating with cloud storage solutions, facilitating scalable data handling for large datasets.

The *Tools* layer (depicted in the right box in the architecture diagram) consists of (external) software that interacts with the `infDB` Services to process data or generate insights. This modular approach allows users to chain different tools into custom workflows.

- **Standardized Integration:** Tools interact with the core database exclusively through open interfaces (SQL or REST APIs), ensuring that the underlying data schema remains consistent regardless of the analysis tool used.
- **pyinfdb:** To facilitate the development of custom tools, the platform provides the `pyinfdb` Python package. This library abstracts database connections, logging, and configuration management, allowing researchers to rapidly develop Python-based analysis scripts that integrate seamlessly with the infDB ecosystem.
- **Extensible Ecosystem:** Because the architecture relies on standard interfaces, users can integrate existing third-party simulation software or develop proprietary tools that plug into the `infDB` backend without modifying the core services.

Platform management is simplified through provided Bash scripts (e.g., for startup and teardown) and environment variable configurations (`.env`), ensuring that the entire infrastructure can be deployed or replicated with minimal effort.

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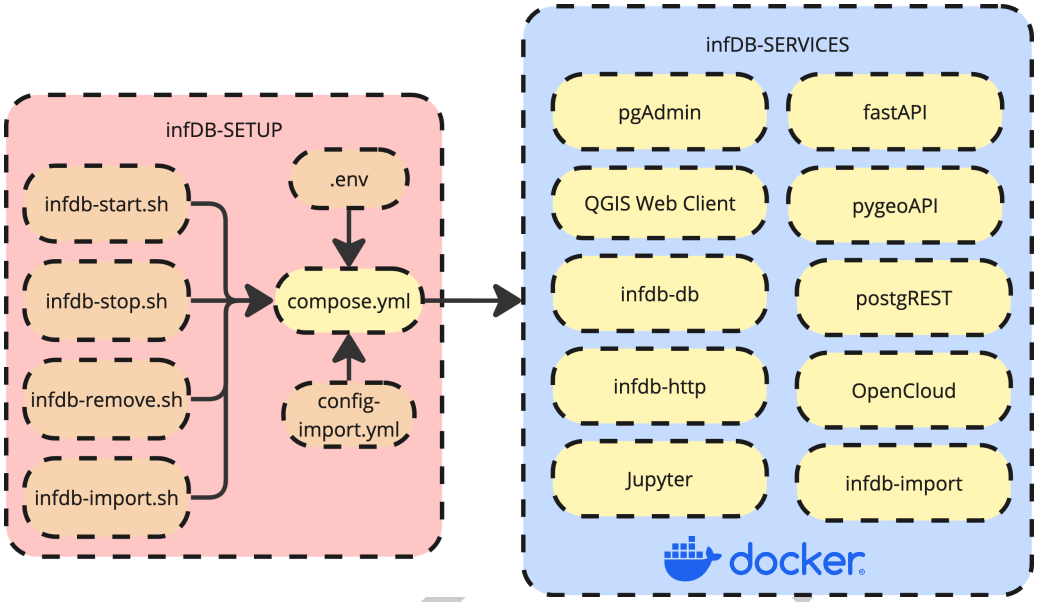

Figure 2: infDB - Architecture

Research Relevance

The major research impact of `infDB` is the standardization of energy system analysis workflows. By decoupling data management from modeling logic, it allows for:

- **Reproducibility:** The containerized approach ensures that an analysis environment can be perfectly replicated on any machine.
- **Longitudinal Analysis:** Automated importers allow researchers to regularly update their datasets as new open data becomes available, transforming static "heat plans" into dynamic monitoring systems.
- **Algorithmic Development:** Researchers can focus on developing algorithms (e.g., for district heating expansion) relying on a stable database schema, rather than adapting algorithms to shifting input formats.

Applications

A key application example is calculating linear heat density for municipal heat planning (KWP). Using `infDB`, researchers and planners can immediately leverage preprocessed, enriched open building data (LOD2) combined with statistical census data to compute building-level heat demand. This approach dramatically reduces computational overhead and development effort compared to traditional file-based GIS workflows, while providing reproducible, auditable results that can be continuously updated as new data becomes available.

AI usage disclosure

In the development of this work, GitHub Copilot, ChatGPT, and Gemini were used. Github Copilot assisted in code generation by code suggestions and completion tasks, while ChatGPT and Gemini assisted in drafting and refining textual content.

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